

Efficient SWIR PbS QD Photodetector Based on A Hot Spin-coating Method

Lei Rao

College of Integrated Circuits and
Optoelectronic Chips
Shenzhen Technology University
Shenzhen, China
202201102083@stumail.sztu.edu.cn

Weichao Wang

School of Materials Science and
Engineering
Hubei University
Wuhan, China
weichaowang@stu.hubu.edu.cn

Xi Chen

College of Integrated Circuits and
Optoelectronic Chips
Shenzhen Technology University
Shenzhen, China
202100102062@stumail.sztu.edu.cn

Jiankai Wang

College of Integrated Circuits and
Optoelectronic Chips
Shenzhen Technology University
Shenzhen, China
202100601050@stumail.sztu.edu.cn

Wei Chen

College of Engineering Physics
Shenzhen Technology University
Shenzhen, China
chenwei@sztu.edu.cn

Haodong Tang

College of Integrated Circuits and
Optoelectronic Chips
Shenzhen Technology University
Shenzhen, China
tanghaodong@sztu.edu.cn

Abstract—PbS quantum dots are promising materials for photodetectors due to their tunable bandgap and broad absorption spectrum. Traditional fabrication methods often result in film inhomogeneity due to uneven solvent evaporation. We introduce a hot spin-coating method that heats the substrate during the spin-coating process, accelerating solvent evaporation and improving film quality. Photoconductors prepared using this method show superior performance compared to those made with conventional techniques.

Keywords—Quantum dots, hot spin-coating, photodetectors

I. INTRODUCTION

PbS quantum dots have received significant attention in recent years as one of the most promising emerging materials for photodetectors with a wide tunable bandgap, absorption spectra spanning the visible to short-wave infrared region, and compatibility with a wide range of substrates[1]. The conventional approach to PbS quantum dot photoconductors requires an annealing operation to remove the solvent after a spin-coating process to form a thin film. It has been learnt that the rate of solvent evaporation affects the quality of the film, that is the structure formed by the bonding of the quantum dots, and that such fabrication is prone to film inhomogeneity, which affects the combination of the PbS quantum dots and hence their photoelectric effect[2].

In this work, thin film preparation was accomplished using a hot spin-coating method, that is, a spin-coater that can be temperature-controlled and used to heat the substrate during the spin-coating process. The high temperature treatment accelerates the evaporation of solvents during the spin-coating process and improves the quality of the film formation, and the results show that the photoconductors prepared by hot spin-coating are better than the devices prepared by the conventional method in terms of the characterization of several quality factors. The aim of this work is to provide a more efficient method for the preparation of PbS quantum dot films, which will provide a new idea for the preparation of future photodetectors.

II. EXPERIMENTAL SETUP

A. Preparation of PbS QDs

The preparation and passivation of PbS quantum dots involved a series of fine chemical processing steps. Firstly, PbS quantum dots with large diameters were synthesized by the conventional Hines method (Fig. 1.a)[3]. Next, a basic ligand exchange was performed to transfer the quantum dots from oleic acid to a mixed solution containing PbBr₂ and PbI₂ to initially passivate the quantum dot surface. Then, dual passivation of the quantum dots was achieved by secondary ligand exchange and butylamine stabilization in lower concentrations of PbBr₂ and PbI₂ solutions, which effectively improved the surface chemical state and charge transport properties of the quantum dots. Finally, these optimally passivated quantum dots were re-dispersed and spin-coated on the electron transport layer, which was subsequently annealed to form a homogeneous and powerful photoactive layer, providing the necessary optoelectronic properties for high-performance SWIR photodetectors[4].

B. Thin Film Fabrication of PbS QDs

A hot spin-coating process is performed to fabricate the PbS QD film. The process sketch is shown in Figure 1.(b, c). A LEBO IC5000-S spin-coater is used to control the temperature of substrate during the spin-coating process. The hot spin-coated film was prepared under the temperature of 85°C, and another group of film was prepared under room temperature (RT). The spin-coating process parameters of the two films are the same, adopting a two-stage method. Firstly, the rotational speed was set at 1200 r/min, the acceleration was 600 r/s, and the spin-coating process was followed for 4 s. Then, the same rotational speed and acceleration were adopted as the benchmark, and the final rotational speed was set at 2400 r/min, and the spin-coating process was carried out for 40 s. The spin-coated devices were annealed for 10 min at the end of the spin-coating process.

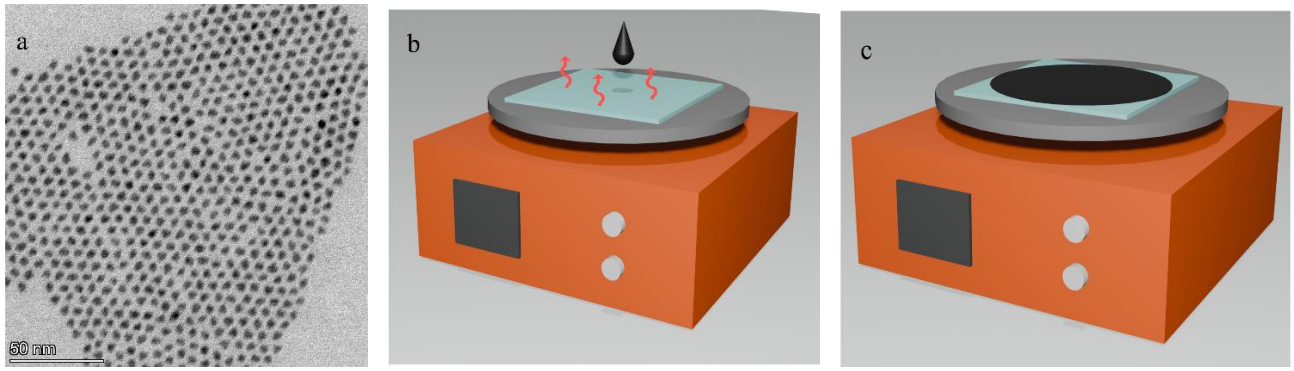


Fig. 1. a) TEM image of PbS quantum dots. b) Schematic diagram of the hot spin-coating process, the substrate is kept at a constant temperature during the spin-coating process. c) Schematic diagram after spin-coating.

C. Characterization of PbS QDs Photodetectors

The photodetectors are characterized using the ENLITECH-Conductor test box, which was interfaced with a Keithley 4200-SCS for precise electrical measurements. The photodetectors were exposed to light emitted from a Xenon lamp (ENTITECH-75W-ZK1071) to evaluate their light response characteristics. Light intensity during the experiments was continuously monitored using a Thorlabs PM 400 light meter and additional detectors. To assess the response characteristics of the photodetectors, a nanosecond laser (MDL-NS-1310) was employed. The current signal of photodetectors is amplified and converted to voltage signal by a low-noise amplifier (SR570). The amplified voltage signal is then captured and analyzed used an oscilloscope (MSO54B).

III. RESULT AND DISCUSSION

A. Current-voltage Characterization

Figure 2 depicts the current-voltage characteristics of the PbS quantum dot photodetectors under different preparation conditions, both in the dark and under light irradiation at a short-wave infrared wavelength of 1310 nm with an optical power of 6.6 mW. It is evident that the device prepared by hot spin-coating (red line) exhibits significantly better performance compared to the device prepared at room temperature (black line). Specifically, under conditions from -2 V to +2 V, the hot spin-coated device shows a 157% higher dark current, a 110% higher photocurrent, and a 109% higher photogenerated current. This device also displays a higher current-voltage response than the room temperature spin-coated device. This improvement may be attributed to the thermal evaporation of the solvent during the hot spin-coating process, resulting in a denser PbS quantum dot film. This density enhances carrier mobility and increases transport speed under the same bias voltage in dark conditions. Under light irradiation, the photocurrent of the hot spin-coated device is several times greater than that of the room-temperature device, due to its increased sensitivity to light and relatively higher carrier mobility[5].

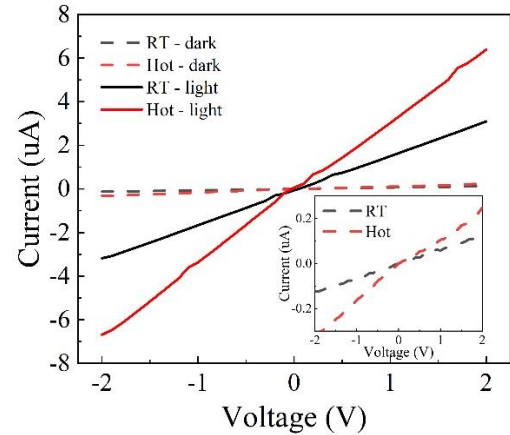


Fig. 2. Relationship between current and voltage for hot spin-coated devices and room temperature spin-coated devices.

B. Absorbance and PL Spectra

Figure 3 shows the absorbance and photoluminescence spectra of this photodetector from 700 to 1700 nm. The absorption peak of the PbS QD occurs at 1341 nm. Additionally, the PbS QD exhibits a photoluminescence intensity peak at 1560 nm. These features indicate that the device has strong infrared detection performance in the short-wave infrared region.

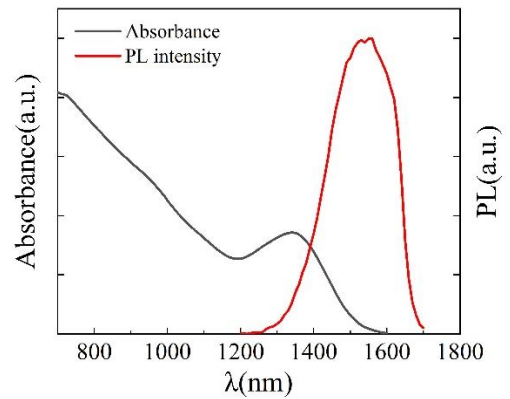


Fig. 3. The Absorption and PL spectra of PbS QDs.

C. Responsivity of Devices

Responsivity ($R(\lambda)$) is one of the most important parameters describing the quality factor of a photodetector and is defined as the ratio of the generated photocurrent or voltage to the incident optical power falling on the detector. In this work the responsivity of the devices is calculated by:

$$R(\lambda) = \frac{I_{ph}(\lambda)}{P_{in}(\lambda)} = \frac{|I_p(\lambda) - I_d(\lambda)|}{P_{in}(\lambda)} \quad (1)$$

where $I_{ph}(\lambda)$ is the photogenerated current value, that is, the absolute value of the photocurrent $I_p(\lambda)$ minus the dark current $I_d(\lambda)$, and $p_{in}(\lambda)$ is the optical power of the incident light in the working area of the device[6].

TABLE I. CURRENT AND RESPONSIVITY OF RT AND HOT SPIN-COATED DEVICES.

Device	$I_d(\lambda)$ [mA]	$I_p(\lambda)$ [mA]	$I_{ph}(\lambda)$ [μ A]	$R(\lambda)$ [A/W]
RT	1.08	1.20	121.8	4.95
Hot	1.48	1.75	270.3	11.03

Table 1 demonstrates the numerical values of the responsivity of the devices of two different processes at a wavelength of 1310 nm, a light power density of 2.90 mW/cm², and a bias voltage of 20 V. Only through numerical observation, it can be seen that the responsivity of the devices prepared by the hot spin-coating process is significantly higher than that of the room-temperature spin-coated devices.

D. Response Time And -3dB Bandwidth

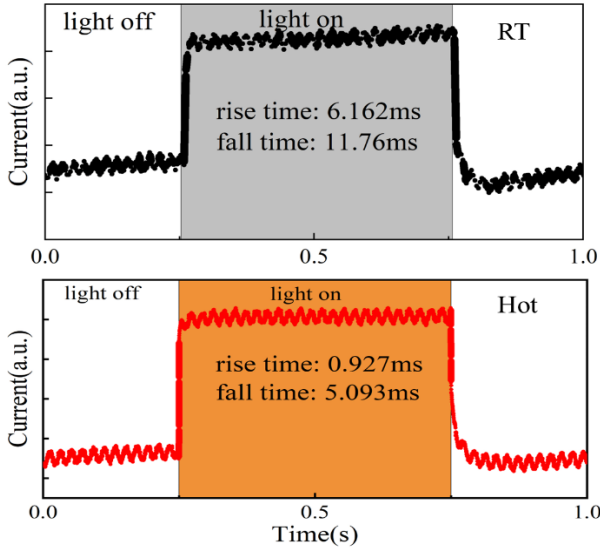


Fig. 4. Response times of photodetectors prepared by two different processes.

In general, the shorter the response time of a photodetector, the faster it responds to changes in the optical signal. Two specific parameters used to evaluate the speed of a photodetector are the rise time (t_r) and the fall time (t_f), defined as the time required for the output signal of the photodetector to rise from 10% to 90% of its final value, and the time required to fall from 90% to 10% respectively[2]. The conventional view holds that the response time includes the

time required for excess carrier recombination and the removal of shallow and deep defects[7]. At a power of 4.8 mW/cm², a frequency of 1 Hz, a duty cycle of 50%, and a wavelength of 1310 nm, the rise time of the room-temperature spin-coated device is 6.162 ms, and the fall time is 11.76 ms. In contrast, the rise time of the hot spin-coated device is 0.928 ms, and the fall time is 5.093 ms; details are provided in Figure 4. Another parameter that can reflect the response speed of photodetectors to changes in the input signal is the 3 dB bandwidth, f_c , which can be used to calculate the response time of the photodetector[1,8]. The calculation formula is:

$$f_c = \frac{2.2}{2\pi t_r} = \frac{0.35}{t_r} \quad (2)$$

The -3 dB bandwidth are calculated as 57 kHz and 377 kHz for the RT and Hot spin-coated devices, respectively. Which further illustrating the superiority of the hot spin-coated device.

IV. CONCLUSION

In this work, we prepared photoconductors using hot spin-coating and room temperature spin-coating methods, and comprehensively evaluated the performance of the devices prepared by these two methods. This evaluation included comparisons of current-voltage characteristics, absorptivity and photoluminescence spectral characteristics, responsivity under a specific bias voltage, and response times. The results demonstrate that the photodetectors fabricated by the hot spin-coating method are superior to those prepared by the room-temperature method in many aspects. The successful application of this technology provides new ideas and methods for the development of high-efficiency photodetectors in the future.

ACKNOWLEDGMENT

This work is supported by Guangdong University Young Innovative Talents Program Project 2023 (No. 2023KQNCX068), Natural Science Foundation of Top Talent of SZTU(No. GDRC202345, No. GDRC202340), National Natural Science Foundation of China (No. 12204318), Guangdong Basic and Applied Basic Research Foundation (No. 2021A1515110535), Shenzhen Science and Technology Program (No. RCYX20221008092908030).

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